Supply Chain Performance Analysis

DataCo Smart Supply Chain Dataset — End-to-End SQL Analytics Project

Madhavan Padmanaban

MSc Operations and Supply Chain Management — University of Liverpool

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Tools: SQL Server • Python (pandas) • Tableau • GitHub

Dataset: DataCo Smart Supply Chain (Kaggle) • 180,519 rows • 46 columns after cleaning

1. Executive Summary

This report presents a comprehensive supply chain performance analysis conducted on the DataCo Smart Supply Chain dataset, a publicly available Kaggle dataset containing 180,519 transactional order records spanning January 2015 to February 2018. The dataset covers 50 product categories, 11 departments, 3 customer segments, and 5 global markets across Latin America, Europe, the Pacific Asia region, the United States, and Africa.

The project followed a full end-to-end analytics workflow: raw data ingestion, Python-based cleaning using pandas, structured SQL analysis in SQL Server, and interactive dashboard development in Tableau. Eleven SQL queries were written and executed covering category and department profitability, delivery performance, demand trends, order status, customer behaviour, ABC inventory classification, and RFM customer segmentation.

The analysis reveals that revenue is heavily concentrated in a small number of product categories, late delivery affects the majority of orders across all regions and shipping modes, and customer purchasing behaviour is largely homogeneous across segments — presenting both operational risks and strategic opportunities for improvement.

2. Key Findings

The following findings represent the most significant outcomes of the eleven SQL analyses. These conclusions are supported by detailed query results in Section 6.

Revenue concentration — Seven product categories generate 77% of total revenue. Fishing alone contributes 18.8% of all sales at $6.9M — a classic Pareto distribution with direct implications for inventory prioritisation.

Late delivery crisis — 54.8% of all orders — 98,977 out of 180,519 — are classified as late delivery. This is consistent across all regions and markets, indicating a systemic issue rather than a region-specific one.

Shipping mode paradox — First Class shipping has a 95.32% late delivery rate, higher than Standard Class at 38.07%. This counterintuitive result suggests scheduling parameters are miscalibrated rather than genuine operational failure.

Order completion gap — Only 32.96% of orders reach COMPLETE status. A further 22.07% remain in PENDING_PAYMENT and 2.25% are flagged as SUSPECTED_FRAUD — representing approximately 7,754 orders at risk of revenue loss.

Revenue vs profitability mismatch — The top customer by sales value (Mary Smith, $10,524) generates a net loss of $866 — demonstrating that revenue alone is a misleading measure of customer value without profitability analysis.

Department imbalance — Fan Shop generates $17.1M in sales — more than double the second-ranked Apparel department. Book Shop generates only $12,587 with the lowest profit margin in the business at 7.91%.

Segment homogeneity — All three customer segments show almost identical profit margins (10.59–10.86%) and late delivery rates (54.73–55.23%), indicating no differentiated service or pricing strategy exists across Consumer, Corporate, and Home Office buyers.

Long tail inefficiency — 33 of 50 product categories (Class C) generate only 5% of total revenue. The bottom five categories combined generate under $40,000 in total sales — representing SKU complexity without meaningful commercial return.

3. Business Recommendations

3.1 Inventory Management

Prioritise Class A categories: Implement dedicated safety stock, daily monitoring, and preferred supplier agreements for the 7 high-value categories — Fishing, Cleats, Camping and Hiking, Cardio Equipment, Women's Apparel, Water Sports, and Men's Footwear.

Review Class C long tail: Evaluate the bottom 5 categories — CDs ($3,060), Toys ($6,105), Golf Bags ($10,369), Soccer ($26,477), and Basketball ($27,099) — for potential discontinuation to simplify the supply chain.

Apply JIT to Class C items: Reduce working capital tied up in low-value inventory by switching Class C categories to Just-in-Time replenishment rather than holding safety stock.

3.2 Delivery and Logistics

Audit First Class SLA parameters: A 95.32% late delivery rate on a premium service is commercially unacceptable. An immediate audit of scheduled delivery time assumptions for First Class orders is required — the data suggests over-promising rather than under-delivering.

Recalibrate all delivery scheduling: Advance shipping orders arrive 1.5 days early on average, suggesting the scheduling model systematically overestimates delivery times. Recalibrating expected delivery dates would reduce apparent late delivery rates without any operational change.

Regional carrier review: Central Africa (57.96%) and South Asia (56.27%) carry the highest late delivery risk — carrier contracts and route planning in these markets should be reviewed.

3.3 Customer Strategy

Adopt profitability-adjusted customer scoring: Replace revenue-only customer rankings with profit-adjusted scoring. Top revenue customers generating net losses (e.g. Mary Smith) should be subject to discount review and pricing adjustment.

Differentiate service by segment: Corporate and Home Office segments warrant premium service level agreements and potentially higher pricing in exchange for more predictable order volumes — currently they receive identical treatment to Consumer buyers.

Re-engage At Risk RFM customers: Customers classified as At Risk under the RFM model were once frequent buyers but have become inactive. Targeted re-engagement campaigns with personalised offers should be prioritised before these customers are lost permanently.

3.4 Order Management

Resolve PENDING_PAYMENT backlog: 22.07% of orders — over 39,000 — are stuck in PENDING_PAYMENT. An automated payment chasing workflow would convert a significant proportion into completed revenue.

Strengthen fraud controls: 4,062 orders (2.25%) are flagged as SUSPECTED_FRAUD. A structured fraud review process with clearer escalation rules would reduce revenue leakage and operational cost.

4. Dataset and Methodology

4.1 Dataset Description

Attribute	Detail
Source	Kaggle — DataCo Smart Supply Chain for Big Data Analysis
Authors	Fabian Constante, Fernando Levano, Katerine Ogando
Raw rows	180,519
Raw columns	53
Cleaned columns	46
Date range	January 2015 — February 2018
Markets covered	Europe, LATAM, Pacific Asia, USCA, Africa
Customer segments	Consumer, Corporate, Home Office
Product categories	50
Departments	11
Shipping modes	First Class, Second Class, Same Day, Standard Class

4.2 Python Data Cleaning

The raw dataset was cleaned using Python and the pandas library in Jupyter Notebook before loading into SQL Server. The following steps were performed in sequence:

Step	Code	Action
1	pd.read_csv(..., encoding='latin-1')	Load CSV with latin-1 encoding to handle special characters in customer city and country names
2	df.shape	Verified 180,519 rows and 53 columns loaded correctly
3	df.head()	Visual inspection of first 5 rows to understand data structure
4	df.isnull().sum()	Identified nulls: Product Description (180,519 — 100% null), Order Zipcode (155,679 null), Customer Zipcode (3 null), Customer Lname (8 null)
5	df.columns.str.lower().str.replace()	Step 1: converted column names to lowercase and replaced spaces with underscores. Step 2: removed brackets from column names
6	df.drop(columns=[...])	Removed 7 columns: product_description (100% null), product_image (URLs only), customer_password, customer_email, customer_street, customer_zipcode, order_zipcode
7	df.rename(columns={...})	Renamed order_date_dateorders to order_date and shipping_date_dateorders to shipping_date for cleaner SQL queries
8	df['customer_lname'].fillna('Unknown')	Filled 8 null values in customer_lname to prevent null string concatenation in SQL name fields
9	df.to_csv(..., index=False)	Exported clean dataset as dataco_cleaned.csv — 180,519 rows, 46 columns, zero nulls remaining
10	SQL Server Import	order_date and shipping_date auto-detected as datetime2(7). late_delivery_risk imported as int. Full datetime and SUM support confirmed — no conversion workarounds needed.

After cleaning, the CSV was imported into SQL Server using the Import Flat File wizard. SQL Server automatically detected and imported order_date and shipping_date as datetime2(7) — full datetime support confirmed with no manual conversion required. The late_delivery_risk column was imported as int, enabling SUM and AVG aggregations directly in all queries without any conversion workaround.

4.3 SQL Approach

Eleven analytical queries were written in SQL Server Management Studio. The queries progress from basic GROUP BY aggregations through to multi-CTE window function analysis and RFM scoring. Two reusable views were created — vw_rfm and vw_abc — and combined into a master view vw_supply_chain_master for Tableau connectivity.

SQL Technique	Queries Used In	Purpose
GROUP BY + Aggregates (SUM, AVG, COUNT)	1, 3, 4, 6, 7, 10	Summarise by category, region, department, segment
CTE (Common Table Expression)	2, 5, 7, 8, 9, 10, 11	Break complex logic into readable named steps
Window Function — SUM OVER	7, 9	Running totals and percentage of total without collapsing rows
Window Function — LAG	5	Month-over-month sales growth comparison
Window Function — RANK	6	Department profitability ranking
Window Function — NTILE(5)	11	RFM scoring — divide customers into 5 equal buckets
CASE Statement	2, 9, 11	Delivery classification, ABC class assignment, RFM segmentation
DATEDIFF	11	Calculate recency — days since customer last ordered
COUNT DISTINCT	1, 4, 10, 11	Count unique orders and customers to avoid row duplication
Correlated Subquery	11	Reference latest date in dataset for recency calculation
CREATE VIEW + LEFT JOIN	Master view	Combine all columns with RFM and ABC as reusable Tableau source

5. Tableau Dashboards

Two interactive dashboards were built in Tableau Public using the cleaned supply chain CSV as the data source, enriched with customer RFM segments and product ABC classifications joined from separate CSV exports.

Dashboard	KPIs	Charts	Filters
Overview	Total Orders, Total Sales, Total Profit, Profit Margin %	Monthly orders bar chart, annual sales trend line chart, geographic sales distribution map	Year, Month
Product Overview	Category, ABC Class, Items Sold, Revenue — filtered by product search	Top 10 categories by sales (horizontal bar), Top 10 products by sales (horizontal bar)	Product Name search

6. Analysis and Findings

This section presents the results of all eleven SQL queries with supporting data tables and analytical interpretation. Queries are presented in the order they were developed, progressing from foundational aggregations to advanced analytical techniques.

6.1 Sales and Profit by Category

Identified total orders, sales, profit, and margin for each product category to understand revenue and profitability concentration.

Category	Total Orders	Total Sales ($)	Total Profit ($)	Margin %
Fishing	15,164	6,929,654	756,221	10.91
Cleats	20,386	4,431,943	494,637	11.16
Camping & Hiking	12,299	4,118,426	427,456	10.38
Cardio Equipment	11,355	3,694,843	383,011	10.37
Women's Apparel	17,869	3,147,800	350,421	11.13
Water Sports	13,758	3,113,844	325,147	10.44
Men's Footwear	18,783	2,891,758	311,903	10.79
... (43 more categories)	—	—	—	—

> *Insight: Fishing alone generates $6.9M — 18.8% of all revenue — from just 15,164 orders. The top 7 categories collectively drive 77% of total sales. Profit margins are remarkably consistent at 10–11% across most categories, suggesting standardised pricing. The exception is Garden at 12.97%, the highest margin category, presenting a low-risk volume growth opportunity.*

6.2 Delivery Performance by Shipping Mode

Used a CTE and CASE statement to calculate late delivery rates and average delay days per shipping mode.

Shipping Mode	Total Orders	Late Orders	Late Delivery %	Avg Delay (days)
First Class	27,814	26,513	95.32%	1.0
Second Class	35,216	26,987	76.63%	1.99
Same Day	9,737	4,454	45.74%	0.48
Standard Class	107,752	41,023	38.07%	0.0

> *Insight: First Class has a 95.32% late delivery rate — higher than Standard Class. This is counterintuitive and almost certainly reflects over-optimistic SLA scheduling rather than genuine operational failure. The average delay for First Class is only 1 day, which suggests scheduled delivery windows are set too aggressively. Recalibrating scheduled dates alone would dramatically improve apparent performance without any logistical change.*

6.3 Late Delivery Risk by Region

Calculated the proportion of orders flagged as late delivery risk by order region. As late_delivery_risk was imported as int, SUM aggregation works directly without any data type conversion.

Region	Total Orders	Late Risk Orders	Late Risk %
Central Africa	1,677	972	57.96%
South Asia	7,731	4,350	56.27%
East Africa	1,852	1,036	55.94%
Western Europe	27,109	15,140	55.85%
South of USA	4,045	2,256	55.77%
Standard Class	107,752	41,023	38.07%

> *Insight: Late delivery risk is high across every region — ranging from 54% to 58% with no region performing well. The most significant finding is that Western Europe (27,109 orders) carries 55.85% late risk despite being a well-developed logistics market. This confirms the issue is systemic across the entire supply chain rather than driven by challenging geographies.*

6.4 Top 10 Customers by Total Sales

Identified the ten highest-value customers by total sales, including customer segment, country, order count, and profit contribution.

Customer	Segment	Country	Orders	Total Sales ($)	Total Profit ($)
Mary Smith	Corporate	EE.UU.	13	10,524	-866
Mary Patterson	Consumer	EE.UU.	11	9,299	1,347
Mary Duncan	Corporate	Puerto Rico	11	9,296	1,496
Betty Phillips	Consumer	Puerto Rico	11	9,224	2,197
Betty Spears	Consumer	EE.UU.	11	9,131	2,442

> *Insight: The top revenue customer (Mary Smith, $10,524) generates a net loss of $866 — a critical finding that demonstrates the risk of measuring customer value by sales alone. Heavy discounting on large corporate orders can erode margins entirely. Betty Spears and Betty Phillips both generate comparable sales with significantly higher profit, making them more genuinely valuable despite lower headline revenue. Customer profitability scoring should replace revenue-only rankings.*

6.5 Monthly Sales Trend with MoM Growth

Used the LAG window function to calculate month-over-month sales growth percentage across the full dataset date range.

Year	Month	Total Sales ($)	Total Profit ($)	Total Orders	MoM Growth %
2015	1	1,051,590	111,661	1,787	NULL
2015	2	927,010	99,141	1,585	-11.85%
2015	3	1,051,254	113,779	1,781	+13.40%
2015	4	1,014,463	108,084	1,710	-3.50%
2016	1	1,046,308	106,781	1,772	-1.09%
2016	2	968,543	86,810	1,650	-7.43%

> *Insight: Monthly sales are stable between $927K and $1.05M across 2015–2017 with no strong seasonal trend. MoM fluctuations of -11% to +13% represent normal demand variation. This stability is a positive indicator for demand forecasting — simple moving average models would produce reliable inventory planning outputs. The 2018 data shows only $331,650 in January, reflecting a partial year dataset rather than a revenue decline.*

6.6 Department Profitability Ranking

Used RANK() window function alongside GROUP BY aggregations to rank all 11 departments by total profit, avoiding the need for a separate CTE.

Rank	Department	Total Orders	Total Sales ($)	Total Profit ($)	Avg Margin %
1	Fan Shop	41,303	17,113,871	1,834,155	12.0%
2	Apparel	35,190	7,976,255	881,883	12.28%
3	Golf	25,889	4,609,028	497,524	11.88%
4	Footwear	13,009	4,006,499	410,222	11.92%
9	Health and Beauty	362	106,080	9,494	9.56%
10	Pet Shop	492	41,524	3,590	9.41%
11	Book Shop	405	12,587	883	7.91%

> *Insight: Fan Shop dominates at $17.1M in sales — more than double Apparel in second place. Despite having the second-lowest average margin (12%), its volume advantage is overwhelming. Book Shop generates only $12,587 in total revenue at the worst margin in the business (7.91%). Given its minimal contribution and below-average margin it is a candidate for discontinuation or consolidation with a related department.*

6.7 Order Status Breakdown

Used SUM() OVER () as a window function to calculate the percentage of total for each order status without a separate aggregation CTE.

Order Status	Count	% of Total
COMPLETE	59,491	32.96%

PENDING_PAYMENT	39,832	22.07%
PROCESSING	21,902	12.13%
PENDING	20,227	11.20%
CLOSED	19,616	10.87%
ON_HOLD	9,804	5.43%
SUSPECTED_FRAUD	4,062	2.25%
CANCELED	3,692	2.05%
PAYMENT_REVIEW	1,893	1.05%

> *Insight: Only 32.96% of orders reach COMPLETE status. With 22.07% stuck in PENDING_PAYMENT and 2.25% flagged as SUSPECTED_FRAUD, approximately 7,754 orders are at risk of never generating revenue. Combined with 2.05% CANCELED, nearly 4.3% of all order attempts result in no revenue — an operational and commercial problem that requires structured intervention.*

6.8 Shipping Delay Analysis

Used a CTE to calculate delay per row (actual days minus scheduled days) before aggregating. Days_for_shipping_real and Days_for_shipment_scheduled are tinyint columns — multiplying by 1 before subtraction promotes them to int, preventing arithmetic overflow when the result is negative.

Delivery Status	Total Orders	Avg Delay (days)	Max Delay	Min Delay
Late delivery	98,977	1.62	4	1
Shipping canceled	7,754	0.57	4	-2
Shipping on time	32,196	0.00	0	0
Advance shipping	41,592	-1.50	-1	-2

> *Insight: Late deliveries average only 1.62 days late — marginal lateness at scale rather than extreme delays. Advance shipping arrives 1.5 days early on average, indicating the scheduling model consistently overestimates delivery times. If scheduled dates were adjusted to be more realistic, a significant portion of current 'late' orders would reclassify as on time. This represents an opportunity to improve KPI performance through data quality rather than operational change.*

6.9 ABC Analysis by Product Category

Used two CTEs — one to aggregate category sales, one to calculate running totals — before classifying each category into A, B, or C using a CASE statement.

ABC Class	No. Categories	Revenue Contribution	Key Categories
A — High Value	7	0% to 77%	Fishing, Cleats, Camping & Hiking, Cardio Equipment, Women's Apparel, Water Sports, Men's Footwear
B — Medium Value	10	77% to 95%	Indoor/Outdoor Games, Shop By Sport, Computers, Electronics, Cameras, Garden, Children's Clothing
C — Low Value	33	95% to 100%	Accessories, Golf Gloves, Music, Consumer Electronics, Health and Beauty, Basketball, Soccer, Toys, CDs

> *Insight: Seven categories generate 77% of revenue while 33 categories generate only 5% combined. The bottom five C-class categories — CDs ($3,060), Toys ($6,105), Golf Bags ($10,369), Soccer ($26,477), Basketball ($27,099) — collectively contribute under $73,000 in total sales across the full dataset period. These categories add SKU complexity, procurement overhead, and warehousing cost that almost certainly exceeds their marginal revenue contribution.*

6.10 Customer Segment Performance

Compared Consumer, Corporate, and Home Office segments across multiple metrics including COUNT DISTINCT for late orders to avoid duplication from one-to-many order item rows.

Segment	Orders	Customers	Total Sales ($)	Avg Order ($)	Margin %	Late Delivery %
Consumer	34,119	10,695	19,095,790	204.22	10.86%	54.73%

Corporate	19,856	6,239	11,168,407	203.84	10.77%	54.75%
Home Office	11,777	3,718	6,520,538	202.34	10.59%	55.23%

> *Insight: All three segments are nearly identical across every metric — average order value differs by just $1.88 between the highest and lowest segment, and profit margins vary by only 0.27 percentage points. This uniformity indicates the business applies no differentiated pricing, discounting, or logistics strategy by segment type. Corporate customers in particular — with their more predictable, higher-volume ordering patterns — would typically warrant premium service levels and higher margins in a mature commercial environment.*

6.11 RFM Customer Segmentation

Used three CTEs to build the RFM model: rfm_base calculates raw R, F, M metrics per customer; rfm_scored applies NTILE(5) window functions to assign scores 1–5; rfm_segmented adds total score and segment classification via CASE statement.

RFM Segment	What it means	Supply Chain Implication
Champion	High recency, frequency, and monetary — best customers	Protect with priority fulfilment and service levels
Loyal	Consistent buyers with good recency and frequency	Upsell and cross-sell — high receptiveness to offers
New Customer	Recent but infrequent — just started buying	Onboard carefully — first experience shapes long-term behaviour
At Risk	Previously frequent but not buying recently	Immediate re-engagement — before they become Lost
Lost	Low on both recency and frequency — disengaged	Win-back campaign or resource reallocation
Potential	Middle ground — has not yet committed to regular buying	Targeted personalised offers to convert to Loyal

> *Insight: The majority of high-spending customers fall into the 'potential' segment — they spend significantly but do not buy frequently enough to qualify as loyal. This indicates a customer base making large infrequent purchases rather than building habitual buying patterns. The business should prioritise converting 'potential' to 'loyal' through repeat purchase incentives, and urgently re-engage 'at risk' customers — previously frequent buyers who have recently gone quiet.*

7. Technical Summary

7.1 Files in This Project

File	Description
datacocleaning.ipynb	Jupyter notebook — full Python data cleaning pipeline
supply_chain_queries.sql	SQL file — 11 analytical queries on dataco_cleaned table
supply_chain_views.sql	SQL file — vw_rfm, vw_abc, and vw_supply_chain_master views
dataco_cleaned.csv	Cleaned dataset — 180,519 rows, 46 columns, exported from Python
customer_seg.csv	RFM segment per customer — exported for Tableau join
product_cat.csv	ABC classification per category — exported for Tableau join

7.2 Known Limitations

• The dataset contains no cost of goods data — profit figures represent order-level profit as calculated within the dataset rather than independently verified margin analysis.

• RFM segmentation thresholds are heuristic — for example r_score >= 4 for Champion classification. A more rigorous approach would use statistical clustering to determine natural segment boundaries from the data itself.

• The 2018 dataset covers January only — a partial year. Year-over-year trend comparisons should treat 2018 figures as indicative rather than comparable to full years 2015–2017.

• Days_for_shipping_real and Days_for_shipment_scheduled are tinyint columns in SQL Server. Subtracting them directly causes arithmetic overflow when the result is negative — resolved by multiplying by 1 before subtraction to promote to int. This would be cleaner if handled during Python cleaning before import.

8. Conclusion

This project demonstrates a complete end-to-end supply chain analytics workflow from raw data through to business intelligence. Starting with 180,519 raw transactional records, the project applied Python-based cleaning, eleven structured SQL analyses using advanced techniques including window functions, CTEs, and RFM scoring, and interactive Tableau dashboards — producing findings with direct operational relevance.

The most significant finding is the systemic late delivery crisis affecting over half of all orders — not driven by genuine operational failure but by miscalibrated scheduling parameters that set unrealistic delivery expectations. Addressing this through data correction rather than operational change represents a quick, low-cost improvement with immediate KPI impact.

The ABC and RFM analyses provide the most strategically valuable outputs — a clear inventory prioritisation framework based on revenue contribution, and a customer behavioural segmentation that identifies where commercial and retention effort will have the greatest return. Together these frameworks give supply chain and commercial teams the analytical foundation needed to make evidence-based decisions about stock management, logistics investment, and customer strategy.

From a technical perspective this project demonstrates practical proficiency across the full modern analytics stack — Python data cleaning with pandas, SQL Server query development covering window functions and multi-CTE logic, Tableau dashboard development, and the application of established supply chain management frameworks to a real-world dataset. It is intended as a practical demonstration of the analytical capabilities directly relevant to supply chain analyst, operations analyst, and PMO roles.